

# Gaussian stress-test methodology

**Goal:** How well does the inverse + surrogate pipeline generalize to Qiskit Metal parameters it has never seen before

Probe the in-between regions of SQuADDS by perturbing held-out test points with controllable Gaussian noise.

## 1. Pick seeds

Choose 30 held-out test samples from SQuADDS and treat each one as the center of a Gaussian ball in Qiskit Metal parameter space.

## 2. Add Gaussian noise

For each parameter  $i$ :

$$p_i^{\text{new}} = p_i^{\text{orig}} + \mathcal{N}(0, \sigma_i)$$

with  $\sigma_i = f \cdot \text{range}(p_i)$ .

Reject samples that leave the box or violate layout constraints.

## 3. Sweep sigma

Sweep  $f$  across

$\{1\%, 2\%, 5\%, 10\%, 20\%\}$   
of each parameter's range.

Draw 6,000 samples per  $\sigma$   
→ 30,000 total.

Surrogate evaluates all 30,000 samples in seconds ( $\sim 5000$  Ansys-hours equivalent).